

Matchmaking Strategies in Two-Sided Markets

Efficiency, Communication Complexity and Near-Optimality

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1 Introduction

Two-sided digital platforms are now a central form of organization in online markets. Their value does not stem from the direct production of a good or service, but from their ability to connect two distinct groups of agents. Typical examples include platforms that connect customers and service providers, clients and freelance workers, hosts and guests, or families and babysitters.

In these markets, the main challenge goes beyond attracting large numbers of users: agents must also be able to identify compatible partners quickly. The platform therefore plays a coordinating role by organizing information, structuring search, and defining the ways in which customers and providers come into contact.

This problem becomes particularly relevant in large online markets. An increase in the number of users expands the set of possible matches, but it can also generate information congestion and make the search process costly and inefficient (Roth and Xing, 1994, 1997; Li and Netessine, 2020). Platform design must therefore facilitate the formation of high-quality matches while limiting communication costs.

Peng Shi’s paper, *Optimal Matchmaking Strategy in Two-Sided Marketplaces*, addresses precisely this issue. The author compares four strategies observed on real platforms: customers search, providers search, both sides search, and centralized matching Shi (2023). These strategies differ in how they distribute search and communication across agents.

The paper’s main contribution is to show that there is no universally superior match-making strategy. The most efficient choice depends on the informational structure of the market, namely on how easily customer and provider preferences can be described through standardized information. When preferences are easy to describe, the platform can play a stronger role in guiding the matching process; when preferences are more idiosyncratic, greater scope must be left to direct search.

From a theoretical perspective, Shi proposes a model based on the assignment game under incomplete information. Each possible customer-provider pair generates a surplus, but the platform does not perfectly know benefits and costs before matching takes place. The efficiency of a strategy is therefore evaluated along two dimensions: the quality of the final outcome and the amount of communication required to reach it.

The objective of this report is to reconstruct and discuss Shi’s model, highlighting the link between the informational structure of the market, communication, and the choice of matchmaking strategy. After introducing two-sided markets, the report examines the theoretical model, approximate stability, the four matchmaking strategies, preference density, and the communication complexity perspective.

The report also includes a numerical experiment in Python. The objective is not to replicate the paper’s proofs in full, but to illustrate numerically how the number of interactions required to reach a good outcome depends on how easy or hard to describe the preferences of the two sides of the market are.

The central argument is that an efficient matchmaking strategy should be consistent with the nature of the information available in the market: the easier preferences are to describe, the greater the role the platform can play; when they are hard to encode, more room must be left for decentralized search.

2 Two-sided markets and market design

2.1 What are two-sided marketplaces?

Two-sided marketplaces are economic environments in which a platform connects two distinct groups of agents. The value of the platform does not derive simply from having many users, but from its ability to facilitate useful interactions between the two sides of the market.

Typical examples include platforms that connect customers and service providers, such as TaskRabbit, Amazon Home Services, or Yelp; clients and freelance workers, such as Upwork; hosts and guests, such as Airbnb; or families and babysitters, such as Care.com. In all these cases, the platform performs an intermediation function: it organizes information, structures search opportunities, and facilitates the formation of matches between compatible agents.

A fundamental characteristic of two-sided markets is the interdependence between the two groups. A customer values the platform if it offers reliable providers suited to the customer's needs; at the same time, a provider participates if the platform offers access to attractive and potentially profitable customers. The platform must therefore coordinate both sides of the market simultaneously.

In Shi's paper, the reference context is that of online platforms that help customers find service providers. The author observes that different platforms adopt different match-making strategies: some allow customers to search for providers, others allow providers to search for customers, others allow both sides to search, while some adopt forms of centralized matching Shi (2023). This variety suggests that no single strategy is suitable for every market.

2.2 The matching problem between customers and providers

The matching problem arises because not all possible customer-provider pairs generate the same value. A customer may obtain a high benefit from one provider and a lower benefit from another. Similarly, a provider may incur different costs depending on the customer served, the type of task, the location, or the time required.

The platform's task is therefore not simply to connect arbitrary users, but to facilitate the formation of compatible pairs. In the setting analyzed by Shi, each possible match is associated with a benefit for the customer and a cost for the provider. A good match is one in which the customer's benefit is high and the provider's cost is relatively low.

However, the platform does not perfectly know this information. The customer has better knowledge of their own preferences, while the provider has better knowledge of their own costs and availability. Matching is therefore both an allocation problem and an informational problem.

The different matchmaking strategies studied in the paper address precisely this issue. A platform may allow customers to search, allow providers to search, permit search on both sides, or intervene directly by recommending a limited number of matches. The central question is therefore under which conditions each strategy is more efficient.

2.3 From matching theory to digital platforms

Matching theory studies how agents belonging to different sets can be assigned to one another in order to obtain a desirable outcome, often one that is stable or efficient. In digital markets, however, the problem is not simply to identify a good matching; it is also necessary to reach it with a limited amount of communication.

This distinction is fundamental. In theory, if the platform perfectly knew all preferences and all costs, it could directly compute the best matching. In practice, collecting all this information may be too costly. Every search, message, proposal, or rejection is a form of communication and requires users' time and attention.

Shi's contribution is to connect the matching problem to communication complexity. A matchmaking strategy is efficient when it produces good matches while reducing the number of interactions required. This approach makes it possible to interpret the market design of digital platforms as a problem of balancing centralization and decentralization (Roth, 2008).

If the information collected by the platform describes users' preferences well, the platform can take a more active role and directly recommend matches. If preferences are instead idiosyncratic (highly specific preferences) and hard to encode, it is preferable to leave more room for search and direct interaction among agents.

The importance of search-process design also emerges from empirical studies. Interventions such as allowing users to signal their preferences, organizing available information more effectively, or introducing recommendation systems can influence the process through which users identify and form a match (Coles et al., 2010; Fradkin, 2017; Horton, 2017).

In summary, two-sided markets require mechanisms capable of coordinating heterogeneous agents while limiting search and communication costs. Shi's paper shows that the choice of matchmaking strategy depends on the informational structure of the market, that is, on the extent to which customer and provider preferences can be described before interaction occurs.

3 Shi's theoretical model

3.1 Assignment game

The theoretical model proposed by Shi is based on the assignment game, introduced by Shapley and Shubik and subsequently developed in the matching literature (Shapley and Shubik, 1971; Roth and Sotomayor, 1990). In the context of the paper, the two sets correspond to customers and service providers on a digital platform Shi (2023).

Let I denote the set of customers and J the set of providers. For simplicity, the paper initially considers the case in which the two sides of the market have the same size:

$$|I| = |J| = n.$$

A matching is represented by a binary matrix x , where $x_{ij} = 1$ if customer i is matched with provider j , and $x_{ij} = 0$ otherwise:

$$x_{ij} = \begin{cases} 1 & \text{if customer } i \text{ is matched with provider } j, \\ 0 & \text{otherwise.} \end{cases}$$

For x to be a valid matching, each customer can be assigned to at most one provider and each provider can serve at most one customer:

$$\sum_{j \in J} x_{ij} \leq 1 \quad \forall i \in I,$$

$$\sum_{i \in I} x_{ij} \leq 1 \quad \forall j \in J.$$

These constraints reflect the one-to-one structure of the basic model. The assignment game is useful because it associates an economic value with every possible customer-provider pair, making it possible to study the surplus generated by matching.

3.2 Benefits, costs, and match surplus

Each possible pair (i, j) , consisting of a customer i and a provider j , is characterized by two main quantities:

- b_{ij} : the benefit that customer i obtains from the service of provider j ;
- c_{ij} : the cost incurred by provider j to serve customer i .

The total surplus generated by the match between i and j is the difference between benefit and cost:

$$S_{ij} = b_{ij} - c_{ij}.$$

A match is therefore more desirable the higher the customer's benefit and the lower the provider's cost. Payment does not change total surplus; it only determines how that surplus is divided between the two agents.

If customer i pays a price p_i to provider j , the customer's surplus is:

$$u_i = b_{ij} - p_i,$$

while the provider's surplus is:

$$v_j = p_i - c_{ij}.$$

Adding the two individual surpluses gives:

$$u_i + v_j = b_{ij} - c_{ij}.$$

This equality shows that the price transfers value from one side of the market to the other without changing the total value generated by the match. For this reason, the paper can analyze markets in which prices are endogenous and help make the match acceptable to both parties.

3.3 Incomplete information and preference distributions

A central element of the model is incomplete information. In real markets, the platform does not perfectly know the benefits and costs associated with every possible customer-provider pair. In addition, some preferences may be hard to communicate in advance through profiles, questionnaires, or filters.

To represent this uncertainty, Shi assumes that benefits and costs are realizations of random variables:

$$\begin{aligned} b_{ij} &\sim B_{ij}, \\ c_{ij} &\sim C_{ij}. \end{aligned}$$

The distribution B_{ij} represents uncertainty about the benefit of customer i from provider j , while C_{ij} represents uncertainty about the cost incurred by provider j when serving customer i .

A market can therefore be represented by the tuple:

$$M = (I, J, B, C),$$

where B is the collection of benefit distributions and C is the collection of cost distributions.

This formulation makes it possible to distinguish between information that is easy to describe and information that is hard to describe. The former can be collected by the platform before matching, for example through profiles, questionnaires, availability, location, or service descriptions. The latter concerns more subjective or idiosyncratic aspects of preferences that emerge only through search and direct communication among agents.

This distinction is fundamental in the paper: the most efficient matchmaking strategy depends precisely on how easy or hard to describe the preferences of the two sides of the market are through standardized information.

3.4 The role of the platform

The platform is responsible for organizing the matchmaking process. It can decide what information to collect, which agents to make visible, who may initiate contact, and whether to leave the process decentralized or intervene through centralized recommendations.

The platform must choose a protocol consistent with the informational structure of the market. If the initial information is highly accurate, the platform can intervene in a more centralized manner. If many preferences are hard to describe, more room must instead be left for search and direct communication.

3.5 Objective: achieving a good outcome with little communication

The central objective of the model is to obtain a good market outcome while reducing the amount of communication required. In a market with complete information, the platform could compute the best matching directly. Under incomplete information, agents must instead communicate information about their benefits, costs, and preferences.

Communication is costly. Every contact, request, rejection, or update requires time and attention. A matchmaking strategy must therefore be evaluated on the basis of both the quality of the final matching and the number of interactions required to achieve it.

The platform's problem can be summarized as follows:

The platform's problem is to design a matchmaking protocol that reaches a stable or nearly stable outcome using little communication.

This formulation combines three dimensions:

- allocative efficiency: the matching must generate value;
- stability: agents must not have strong incentives to deviate;
- informational efficiency: the outcome must be reached with few interactions.

4 Stability and a desirable outcome

4.1 Exact stability and approximate stability

In matching models, stability is a central property: an outcome is stable when there are no pairs of agents that have an incentive to form an alternative match (Gale and Shapley, 1962; Roth and Sotomayor, 1990).

A more flexible notion is adopted, called approximate stability or ε -stability. Under this notion, small opportunities for improvement are tolerated, while deviations large enough to make the formation of an alternative match worthwhile must be excluded.

This formulation makes the concept of stability more suitable for settings in which searching for a new partner and renegotiating entail costs for agents.

4.2 The notion of epsilon-stability

An outcome is represented by the triple (x, u, v) , where x is the matching, u_i is the surplus obtained by customer i , and v_j is the surplus obtained by provider j . For a possible pair (i, j) , the total surplus that the match can generate is:

$$b_{ij} - c_{ij}.$$

An outcome is ε -stable if it satisfies two conditions. The first is the absence of an ε -blocking pair:

$$u_i + v_j + 2\varepsilon \geq b_{ij} - c_{ij} \quad \forall (i, j) \in I \times J.$$

This condition means that no customer-provider pair can improve significantly by deviating from the current matching. The term 2ε represents the pair's total tolerance, accounting for both the customer and the provider.

The second condition is individual rationality:

$$u_i \geq 0 \quad \forall i \in I,$$

$$v_j \geq 0 \quad \forall j \in J.$$

This condition requires that no agent receive negative surplus. Every customer and every provider must therefore be at least weakly better off than under the option of remaining unmatched.

If $\varepsilon = 0$, approximate stability coincides with exact stability. In this case, no pair can obtain any improvement by deviating:

$$u_i + v_j \geq b_{ij} - c_{ij} \quad \forall (i, j) \in I \times J.$$

When $\varepsilon > 0$, the model allows small deviations from the individual optimum, consistently with the idea that agents may tolerate some suboptimality when the cost of searching for alternatives is positive.

4.3 Economic interpretation of epsilon

The parameter ε represents agents' degree of tolerance for an outcome that is not perfectly optimal. Economically, it can be interpreted as a measure of search costs, switching costs, or the uncertainty associated with looking for a new partner.

On a digital platform, changing match is not costless. A customer must search for new alternatives, send requests, and wait for responses; a provider must evaluate new opportunities, estimate costs, and negotiate terms. If the improvement available from a deviation is small, these costs may make it preferable to accept the current matching.

The parameter ε therefore makes it possible to model realistic agent behavior. Users do not necessarily search for the theoretically perfect match, but for a sufficiently good match that can be reached at a limited search cost. For the platform, this means that it is unnecessary to eliminate every minor inefficiency: it is sufficient to reach an outcome that is stable enough while reducing the amount of communication required.

4.4 Connection with social welfare

Approximate stability is also important because it guarantees a good level of social welfare. Social welfare can be interpreted as the total surplus generated by the matching:

$$W(x) = \sum_{i \in I} \sum_{j \in J} (b_{ij} - c_{ij}) x_{ij}.$$

This expression sums the surplus of all pairs that are actually matched. A socially optimal matching would maximize this surplus. However, computing and implementing the optimal matching may require a very large amount of information, because the platform would need to know the benefits and costs associated with every possible customer-provider pair.

The relevant result in the paper is that an ε -stable outcome guarantees welfare close to the optimum. In particular, average surplus per agent is at most ε away from what can be obtained under the best possible matching. Intuitively, if x' is any alternative matching, then:

$$\frac{1}{2n} \left(\sum_{i \in I} u_i + \sum_{j \in J} v_j \right) \geq \frac{1}{2n} \sum_{i \in I} \sum_{j \in J} (b_{ij} - c_{ij}) x'_{ij} - \varepsilon.$$

This relationship shows that approximate stability is both a behavioral condition and an efficiency guarantee. If a platform can reach an ε -stable outcome with little communication, it obtains a matching that is sufficiently stable and close to the social optimum without collecting all the information required to compute the perfect matching.

In summary, the notion of ε -stability connects three central dimensions of the paper: individual stability, social efficiency, and communication efficiency. It allows matchmaking strategies to be evaluated according to both the quality of the final outcome and the amount of information required to reach it.

5 Matchmaking strategies

After defining the model and the notion of a desirable outcome, the paper analyzes four different strategies through which a platform can organize the matchmaking process. The objective is to understand how the search and communication process changes depending on who can search, who can initiate contact, and what role the platform assumes.

The four strategies correspond to arrangements actually observed on digital platforms: customers search, providers search, both sides search, and centralized matching Shi (2023). Each strategy therefore represents a different organization of the process through which information is progressively revealed and used to form matches.

5.1 Customers search

The first strategy is one in which customers search for providers. In this case, the platform provides a browsable directory of service providers. Providers make information available through their profiles, while customers use search tools, filters, and sorting options to identify the alternatives they consider most attractive.

This arrangement is observed, for example, on platforms such as Airbnb, Google Local Finder, and Yelp in its original configuration. The customer compares available providers and independently decides whom to contact; the platform does not directly determine the match, but organizes the information needed to facilitate search.

This strategy is represented by Protocol 1. Each customer progressively evaluates providers on the basis of estimated surplus and contacts the one that appears most advantageous. The interaction reveals new information about the provider's actual cost: in the event of rejection, the customer updates the estimate and continues searching; in the event of acceptance, a match can be formed.

The process is therefore decentralized and sequential: initially incomplete information is progressively updated through interactions between customers and providers.

From a technical perspective, Protocol 1 can be interpreted as an extension under incomplete information of ascending-auction mechanisms used in assignment problems (Demange et al., 1986; Bertsekas, 1988).

5.2 Providers search

The second strategy has a symmetric structure: providers search for customers. The platform collects requests, postings, or descriptions of tasks published by customers and allows providers to identify the opportunities they consider most compatible with their characteristics.

In this case, the customer describes the service required, while providers observe the available requests and decide which customers to contact or which tasks to apply for. The platform therefore facilitates access to and organization of information, but leaves the provider side with the initiative in establishing contact.

In the paper, Protocol 2 is defined as the mirror image of Protocol 1: the roles of customers and providers are exchanged while retaining the same logic of progressive search and information updating Shi (2023).

The fundamental difference from the previous strategy therefore concerns the side of the market on which search activity falls: in the first case, customers progressively select providers, whereas in the second, providers select the opportunities offered by customers.

5.3 Both sides search

The third strategy allows both sides of the market to search for potential partners and initiate contact. The platform does not directly assign matches and does not concentrate search activity on a single side: both customers and providers can observe information, explore the market, and initiate interactions.

This configuration can be observed on platforms such as Care.com and Upwork, where the matchmaking process may depend on characteristics that are hard to represent fully through standardized information.

Protocol 3 extends the customers search mechanism. Before customers begin searching, some providers may directly contact specific customers and communicate additional information about their costs. Customers then proceed with a search similar to that specified in Protocol 1, but using the information revealed in the previous phase.

The main difference from one-sided strategies is therefore that the process of information revelation is distributed across both sides of the market. Contact does not proceed exclusively in one direction, but can be initiated by both customers and providers.

5.4 Centralized matching

The fourth strategy is centralized matching, in which the platform collects information from customers and providers and uses it to directly recommend a limited set of possible matches.

Users therefore do not need to explore the entire directory of potential partners. The platform preselects several alternatives considered plausible, which agents can subsequently accept or reject.

In Shi’s Protocol 4, a set of recommended providers is defined for each customer. The platform checks whether both parties consider the potential match acceptable and, among pairs that mutually accept the recommendation, determines a matching that maximizes the number of matches Shi (2023).

Compared with the previous protocols, communication is therefore more heavily mediated by the platform. In addition, recommendations allow agents to focus their attention on a restricted set of potential partners rather than evaluating the entire market.

5.5 Comparison of the strategies

The four strategies differ mainly in the distribution of search activity and the degree of platform intervention. In the first three strategies, a substantial share of information is revealed through decentralized interactions between customers and providers. Under centralized matching, the platform makes greater use of information available before matching to directly narrow the set of alternatives.

The comparison among these strategies, however, cannot be based solely on the final outcome. The central point of the paper is to determine how much communication is required to reach a satisfactory outcome and how this amount depends on the informational structure of the market.

The relative efficiency of the four protocols depends on the extent to which the preferences of the two sides can be inferred from the information available before interaction. The next chapter formalizes this property through the concept of preference density, introducing the parameters d_I and d_J and using them to identify the conditions under which each strategy provides stronger performance guarantees.

6 Preference density and main result

6.1 Easy and hard to describe preferences

The main result of Shi’s paper is that there is no universally superior matchmaking strategy. The choice among customers search, providers search, both sides search, and centralized matching depends on the informational structure of the market, namely on how easy to describe the preferences of the two sides are through standardized information Shi (2023).

In the paper, a preference is considered easy to describe when it can be approximated with sufficient accuracy using information collected before matching. Such information may come from profiles, questionnaires, filters, availability, location, service characteristics, or users’ initial descriptions. By contrast, a preference is hard to describe when it contains an idiosyncratic component that emerges only through search and direct interaction.

This distinction is important because it determines how much communication is necessary to reach a good outcome. If the initial information is sufficiently informative, the platform can guide matching more efficiently. If the relevant preferences cannot be easily encoded, more room must be left for decentralized search.

6.2 Customer preference density

To formalize this intuition, Shi introduces the concept of preference density. The first parameter is d_I , which measures how easy to describe customer preferences are.

Let $\bar{b}_{ij}$ denote the maximum possible value of benefit b_{ij} , that is, the upper bound of distribution B_{ij} . The parameter d_I is defined as the largest value such that:

$$P(b_{ij} \geq \bar{b}_{ij} - \varepsilon) \geq d_I \quad \forall (i, j) \in I \times J.$$

This condition measures the probability that the customer’s realized benefit is close to the maximum value anticipated by the platform. If d_I is high, the initial information makes it possible to predict reasonably well when a customer can obtain a high benefit from a given provider.

Intuitively, a high value of d_I indicates that customer preferences are relatively easy to describe. This does not mean that all customers have simple preferences, but that the information collected by the platform is sufficient to identify, with high probability, matches that generate high benefit for the customer side.

6.3 Provider preference density

Symmetrically, the parameter d_J measures how easy to describe provider preferences are. In this case, attention is placed on the cost incurred by the provider to serve a given customer.

Let $\underline{c}_{ij}$ denote the minimum possible value of cost c_{ij} , that is, the lower bound of distribution C_{ij} . The parameter d_J is defined as the largest value such that:

$$P(c_{ij} \leq \underline{c}_{ij} + \varepsilon) \geq d_J \quad \forall (i, j) \in I \times J.$$

This condition measures the probability that the provider's realized cost is close to the minimum cost anticipated by the platform. If d_J is high, the platform can predict reasonably well when a provider can serve a customer at low cost.

Intuitively, a high value of d_J indicates that provider preferences or costs are easy to describe. This occurs when the relevant characteristics of the service can be communicated in standardized form and allow the provider, or the platform, to estimate the attractiveness of a given match easily.

6.4 Choosing the strategy based on d_I and d_J

The parameters d_I and d_J link the informational structure of the market to the most efficient matchmaking strategy. The logic is as follows: when the preferences of one side are easy to describe, the other side can search more efficiently because it has sufficient information to guide the search.

If d_J is high, provider preferences are easy to describe. In this case, it is efficient to let customers search, because they can evaluate providers on the basis of reliable information. The most suitable strategy is therefore customers search.

If, instead, d_I is high, customer preferences are easy to describe. Providers can then assess more accurately which customers or requests are compatible with their characteristics. The most suitable strategy is therefore providers search.

When neither d_I nor d_J is high, the preferences of both sides are hard to describe. In this situation, search conducted by only one side can generate many unproductive interactions. It therefore becomes more efficient to allow both sides to search and reveal information through direct contact. The most suitable strategy is therefore both sides search.

Finally, when both d_I and d_J are high, the preferences of both sides are easy to describe. The platform can then use the information collected to directly recommend a small set of matches. The most suitable strategy is therefore centralized matching.

Informational condition	Interpretation	Efficient strategy
d_J high	Provider preferences are easy to describe.	Customers search
d_I high	Customer preferences are easy to describe.	Providers search
d_I low and d_J low	The preferences of both sides are hard to describe.	Both sides search
d_I high and d_J high	The preferences of both sides are easy to describe.	Centralized matching

Table 1: Relationship between preference density and matchmaking strategy

The platform should not choose the matchmaking strategy simply on the basis of market size or the number of users, but primarily on the quality of the information available before matching.

6.5 Interpretation of the results

The preference densities d_I and d_J directly affect search cost. Under customers search, greater ease in describing provider preferences reduces the number of interactions required:

$$\text{interactions in customers search} \approx O\left(\frac{1}{\varepsilon d_J}\right).$$

Symmetrically, under providers search, search cost decreases as d_I increases:

$$\text{interactions in providers search} \approx O\left(\frac{1}{\varepsilon d_I}\right).$$

When both preference densities are low, search conducted by only one side becomes less effective. In this case, both sides search distributes the discovery process across customers and providers and can reduce the overall communication burden.

In the opposite case, with high d_I and d_J , the platform has sufficient information to directly narrow the set of potential partners. Centralized matching can therefore rely on a limited number of recommendations, which in theoretical terms follows the order:

$$\text{recommendations in centralized matching} \approx O\left(\frac{\ln n}{d_I d_J}\right).$$

These relationships describe the theoretical behavior of the protocols and are not operational formulas valid for every platform.

A critical issue concerns the cost required to make preferences easier to describe. A higher preference density may require more detailed profiles, longer questionnaires, or greater information collection by the platform. A reduction in interactions during matching may therefore be accompanied by a higher informational cost in the preceding stage. Overall efficiency also depends on how this information is collected.

7 Communication complexity and near-optimality

7.1 From interactions to communication complexity

In the previous chapters, the efficiency of matchmaking strategies was analyzed mainly through the number of interactions required to reach a good outcome. This measure is useful for studying an individual protocol, but it does not provide a fully homogeneous comparison of strategies that may differ substantially from one another.

An interaction can, in fact, contain very different amounts of information. A simple positive or negative response is not informationally equivalent to communicating an entire preference profile. Similarly, a single interaction with the platform may require a user to transmit information concerning many potential partners.

Shi adopts the communication complexity perspective, an area of information theory that studies the minimum amount of communication required to solve a given problem (Kushilevitz and Nisan, 2006; Shi, 2023). The problem is to obtain an ε -stable outcome with high probability while using as little communication as possible.

Communication is measured in bits, providing a common metric with which to compare different matchmaking protocols. The theoretical question therefore becomes:

How many bits of communication per agent are required, in the worst case, to obtain an ε -stable outcome with high probability?

The use of communication complexity in matching problems predates Shi’s work. Previous studies analyzed the amount of information required to obtain stable matchings, showing that exact stability can require high levels of communication (Segal, 2007; Gonczarowski et al., 2015).

7.2 Lower bound, upper bound, and near-optimality

The analysis is based on a comparison between a lower bound and an upper bound. The lower bound represents the minimum amount of communication that any protocol must necessarily use, in the worst case, to reach the desired outcome. The upper bound represents the maximum cost guaranteed by a specific protocol.

When the upper bound achieved by a protocol matches the lower bound at the same asymptotic order, or differs from it only by a bounded or logarithmic factor, the protocol can be considered near-optimal.

Near-optimality therefore does not imply that a strategy is optimal in an absolute sense. Rather, the result indicates that, within the class of markets considered, there is no matchmaking system capable of substantially reducing the communication required while maintaining the same guarantees on the outcome.

7.3 Theoretical guarantees of the protocols

For a constant value of ε , Shi compares the communication costs of the four protocols with their corresponding theoretical bounds.

For Protocol 1, Customers search, when d_J is bounded below by a positive constant, the lower bound is

$$\Omega(\log n),$$

while the protocol achieves

$$O(\log n).$$

More generally, when d_J can decrease to the order of $1/\sqrt{n}$, the comparison becomes

$$\Omega\left(\frac{1}{d_J}\right) \quad \text{vs.} \quad O\left(\frac{\log n}{d_J}\right).$$

Protocol 2, Providers search, has symmetric results with d_J replaced by d_I .

When no favorable conditions are imposed on the preference densities, there exist markets in which any protocol requires at least

$$\Omega(\sqrt{n})$$

bits of communication per agent. Protocol 3, Both sides search, achieves an upper bound of

$$O(\sqrt{n} \log n).$$

The difference is therefore only a logarithmic factor.

For Protocol 4, Centralized matching, the analysis is conducted within the class of one-round matchmaking protocols, in which each agent communicates without first observing the messages of the others. When the conditions required by the model are satisfied and the product $d_I d_J$ is sufficiently high, the lower bound is

$$\Omega(\log n),$$

while the centralized protocol achieves

$$O(\log n).$$

In this case, the lower and upper bounds therefore have the same asymptotic order.

These results show that the observed cost of a strategy must be evaluated relative to the informational difficulty of the problem. A protocol may require an increasing amount of communication as n grows and still be close to the best theoretically attainable performance if a comparable cost is unavoidable for any protocol.

7.4 From the theoretical result to numerical analysis

Communication complexity therefore provides a theoretical benchmark for evaluating protocol efficiency, but the lower and upper bounds in bits derive from information-theoretic arguments and are not reproduced numerically in this report.

Chapter 8 instead introduces a Monte Carlo experiment that analyzes the problem from a complementary perspective. The numerical analysis observes the number of interactions and matching quality as market size n and preference densities vary, according to configurations specific to the two markets considered. In this way, the experimental analysis makes it possible to observe concretely the relationship between communication cost, the informational structure of the market, and outcome quality.

8 Numerical experimentation

8.1 Objective of the experiment

To complement the theoretical analysis, a numerical project was developed in Python with the objective of observing the behavior of the four matchmaking strategies discussed by Shi in a simulated and controlled environment. The simulation does not attempt to reproduce the information-theoretic proofs of the lower bounds presented in the paper, but to make the main mechanisms underlying the theoretical results observable Shi (2023).

In particular, the experiment studies two dimensions. The first concerns market growth: it observes how the amount of communication required by the protocols changes as the number of customers and providers varies. The second concerns the informational structure of the market: through a sensitivity analysis, the preference density parameters d_I and d_J are varied to observe how the relative performance of the strategies changes.

The objective is therefore not to identify a strategy that is always superior to the others, but to test numerically the paper’s central intuition: different protocols can produce outcomes of comparable quality while requiring very different amounts of communication, and the most efficient strategy depends on how easy or hard to describe preferences are on the two sides of the market.

8.2 Project structure

The project was organized by separating simulation logic from result visualization. The backend contains market generation, implementation of the four protocols, metric computation, and execution of the Monte Carlo experiments. The frontend, developed with Streamlit and Plotly, is used exclusively to explore and compare the results obtained.

The main structure of the project is as follows:

Module	Function
<code>config.py</code>	definition of the experiment’s general parameters and the different numerical presets;
<code>markets.py</code>	definition of the two markets, preference densities, and generation of benefit and cost matrices;
<code>protocols.py</code>	implementation of the four matchmaking strategies;
<code>metrics.py</code>	computation of welfare, optimal welfare, match rate, and descriptive statistics;
<code>experiments.py</code>	construction and execution of the Monte Carlo experiment, aggregation, and saving of results;
<code>cli.py</code>	terminal interface for running the simulation;
<code>frontend/app.py</code>	Streamlit dashboard for graphical analysis of results;
<code>tests/</code>	automated tests for checking the implementation.

This separation makes it possible to run the entire experiment without using the graphical interface and makes the process reproducible through a single configuration.

8.3 Generation of simulated markets

The simulation retains the assignment game structure described in the previous chapters. For each possible customer-provider pair (i, j) , a benefit b_{ij} for the customer and a cost c_{ij} for the provider are generated.

To make the preference densities directly controllable, discrete two-point distributions are used:

$$b_{ij} = \begin{cases} 1 & \text{with probability } d_I, \\ 0.6 & \text{with probability } 1 - d_I, \end{cases}$$

and

$$c_{ij} = \begin{cases} 0 & \text{with probability } d_J, \\ 0.4 & \text{with probability } 1 - d_J. \end{cases}$$

The main experiment uses

$$\varepsilon = 0.10.$$

Because $\varepsilon < 0.4$, the benefit is within ε of its upper support only when $b_{ij} = 1$, while the cost is within ε of its lower support only when $c_{ij} = 0$. Consequently,

$$P(b_{ij} \geq 1 - \varepsilon) = d_I$$

and

$$P(c_{ij} \leq 0 + \varepsilon) = d_J.$$

The parameters used in the simulation therefore correspond directly to the preference densities defined in the theoretical model.

The numerical values do not represent empirical estimates obtained from proprietary platform data. Airbnb and Care.com are used as archetypes of two real markets discussed by Shi, while d_I and d_J are experimental calibrations used to represent different informational structures.

8.4 The two markets analyzed

The first configuration represents the short-term rentals market, using Airbnb as a reference. In this case, customer preference density is held constant:

$$d_I = 0.10,$$

while provider preference density is varied:

$$d_J \in \{0.20, 0.40, 0.60, 0.80, 1.00\}.$$

The reference configuration is

$$(d_I, d_J) = (0.10, 0.80).$$

This calibration represents a market in which the relevant characteristics on the provider side are relatively easy to describe, while the customer retains a more idiosyncratic component of preferences.

The corresponding theoretical prediction is therefore favorable to the Customers search strategy.

The second market represents childcare, using Care.com as a reference. In this case, the preference densities of the two sides are varied jointly:

$$d_I = d_J \in \{0.02, 0.05, 0.10, 0.20, 0.40\}.$$

The reference configuration is

$$(d_I, d_J) = (0.02, 0.02).$$

The initial value represents a situation in which both sides have a strong component of preferences that is hard to communicate in advance. In this configuration, the theoretical prediction favors the Both sides search strategy. The progressive increase in d_I and d_J then makes it possible to observe what happens as preferences become progressively easier to describe.

8.5 Implementation of the matchmaking strategies

For each generated market, the four strategies are run on the same benefit matrix B and the same cost matrix C . The comparison is therefore conducted on the same market realization.

Under Customers search, each customer progressively selects the provider that maximizes estimated surplus on the basis of the information available. In the event of rejection, the provider's realized cost is revealed and the estimate is updated. When a provider accepts, the match is formed; if the provider was already temporarily assigned, the previous customer returns to the search process. The protocol also records the number of rejections, acceptances, and reassignments.

Providers search implements the same logic symmetrically. In this case, providers search among customers, using initial benefit estimates and updating them following interactions.

The Both sides search strategy introduces a preliminary phase in which some providers may directly communicate information about their costs. Given the discrete distribution used in the simulation, the parameter z is chosen between two possible levels, 0 and 0.4, by comparing the expected cost of the two modes. After this preliminary phase, the Customers search mechanism is executed using the additional information already revealed. This implementation is a simplified numerical version of Shi's Protocol 3, adapted to the two-point distribution used in the experiment.

Finally, under Centralized matching, the platform selects a limited set of recommended providers for each customer. The number of recommendations per customer is determined by

$$k = \min \left\{ n, \left\lceil \frac{3 \ln n}{d_I d_J} \right\rceil \right\}.$$

A recommendation is considered accepted when both conditions

$$b_{ij} \geq 1 - \varepsilon$$

$$c_{ij} \leq \varepsilon$$

are satisfied. A maximum-cardinality matching is then computed among the accepted pairs. The implementation uses the assignment algorithm available in the SciPy library.

8.6 Monte Carlo design and sensitivity analysis

The experiment was constructed as a single Monte Carlo experiment. A baseline and a sensitivity analysis are therefore not generated separately: for each preference-density configuration, the same market sizes are analyzed, while the baseline is simply one of the points on the experimental grid.

In the main preset, called **standard**, the market sizes considered are

$$n \in \{25, 50, 100, 150, 200\}$$

and 100 Monte Carlo replications are generated for each configuration. Considering two markets, five preference-density levels, and five market sizes gives

$$2 \times 5 \times 5 \times 100 = 5\,000$$

scenario-specific instances. Since each instance is evaluated through four protocols, the complete experiment comprises 20,000 executions of matchmaking strategies.

To reduce noise in the sensitivity analysis, a common random numbers technique is used. For a given market, size n , and replication, different preference-density values use the same random base. The same matrices of uniform random numbers are therefore compared against different thresholds. This allows more direct comparisons across different levels of d_I and d_J .

Similarly, within each configuration, all four protocols are applied to the same realization of B and C . The observed differences among strategies therefore do not depend on different randomly generated markets.

A **quick** preset is also available for rapidly checking that the code functions correctly, as well as a **full** preset intended for possible robustness checks with more replications and larger market sizes.

8.7 Evaluation metrics

The first measure used is the number of interactions per customer:

$$C(x) = \frac{\text{total interactions}}{n}.$$

This variable represents the directly observable communication cost of the strategy.

Matching quality is instead evaluated through social welfare:

$$W(x) = \sum_{i,j} (b_{ij} - c_{ij})x_{ij}.$$

For each market, the maximum welfare attainable under perfect information is also computed:

$$W^* = \max_x \sum_{i,j} (b_{ij} - c_{ij})x_{ij}.$$

The benchmark is calculated by solving a maximum-weight assignment problem. The comparison between the protocol outcome and this benchmark is expressed through the welfare ratio:

$$WR = \frac{W(x)}{W^*}.$$

A value close to one indicates that the strategy produces an outcome very close to the matching that maximizes welfare under complete information.

The welfare regret,

$$1 - WR,$$

and the match rate, namely the share of customers that complete the process with an assigned provider, are also computed.

For each metric, the mean, standard deviation, median, fifth and ninety-fifth percentiles, and 95% confidence intervals are stored.

Finally, a synthetic measure of communication efficiency was introduced. For each replication, only strategies that achieve

$$WR \geq 0.95$$

are considered. Among these, the strategy using the fewest interactions is identified. In the event of a tie, credit is divided equally among the strategies involved. The final share therefore indicates how frequently each protocol is the least costly in terms of communication while simultaneously maintaining a high-quality outcome.

8.8 Interpretation of the results

The results show that strategies with a similar welfare ratio may require very different amounts of communication. For this reason, efficiency must be evaluated by considering outcome quality together with the number of interactions required.

In the reference Airbnb configuration, Customers search reaches welfare close to the benchmark with few interactions per customer, while Providers search requires a higher communication cost. When the parameter z is equal to zero, the implemented version of Both sides search is operationally identical to Customers search. Moreover, as d_J increases while d_I is held fixed, the cost of customer search tends to decrease.

In the reference Care.com market, with $d_I = d_J = 0.02$, Both sides search instead shows the clearest communication advantage. The one-sided strategies require more interactions, while Centralized matching is particularly costly because of the low preference densities.

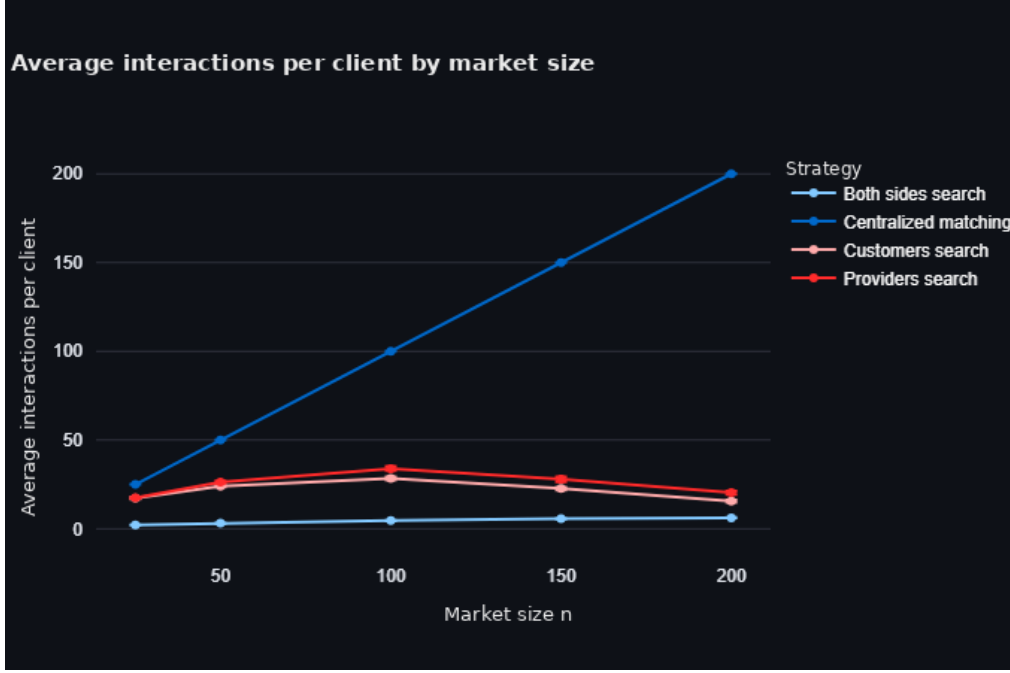


Figure 1: Average interactions per customer in the reference Care.com configuration ($d_I = d_J = 0.02$, $\varepsilon = 0.10$). Each point represents the mean over 100 Monte Carlo replications for the corresponding strategy and market size.

As Figure 1 shows, Both sides search maintains the lowest average number of interactions per customer across all market sizes considered. The result is consistent with a configuration in which the preferences of both sides are hard to describe before interaction.

Centralized matching instead improves as the preference densities increase, because the platform has more useful information for narrowing the set of potential partners.

Overall, the simulation numerically confirms the role of the informational structure of the market: outcomes of similar quality can be reached with very different communication costs depending on the strategy adopted.

8.9 Visualization, reproducibility, and validation

The main experiment was run using the `standard` preset and four parallel workers. This configuration uses 100 Monte Carlo replications for each experimental combination and distributes execution across four parallel processes, reducing computation time without changing the simulation results.

The results are automatically saved in three files. `experiment_raw.csv` contains all individual protocol runs; `experiment_summary.csv` contains aggregate statistics and confidence intervals; `experiment_efficiency.csv` contains the communication-efficiency measure conditional on outcome quality.

The frontend developed in Streamlit allows the same results to be analyzed from three main perspectives. The first shows scaling with respect to n while holding d_I and d_J fixed; the second shows sensitivity with respect to the preference densities while holding market size fixed; the third compares the communication-efficiency share of the four strategies.

Finally, the project includes automated tests to verify several fundamental implementation requirements. In particular, the tests verify that the protocols produce valid one-to-one matchings and that the welfare obtained cannot exceed the benchmark under

perfect information.

The tests also verify that the preference-density grids are constructed correctly and that the common random numbers mechanism produces consistent realizations across the different scenarios.

The separation among data generation, protocols, metrics, experiments, and visualization therefore makes the experiment reproducible and allows individual elements of the model to be modified without altering the entire project structure.

8.10 Limitations of the experiment

The simulation should be interpreted as a numerical and pedagogical extension of the theoretical model, not as an empirical estimate of the actual performance of Airbnb or Care.com. The platforms are used as reference archetypes for representing different informational structures, while the values of d_I and d_J are controlled calibrations.

In addition, benefits and costs follow particularly simple discrete distributions. This choice makes it possible to link the simulation parameters directly to the definition of preference density and clearly isolate the effects being analyzed, but it does not capture all the heterogeneity observable in real markets.

Finally, communication is measured primarily through the number of interactions. The experiment does not attempt to reconstruct numerically the lower and upper bounds in bits studied through communication complexity. Those results remain part of Shi's theoretical analysis, while the numerical project focuses on the observable trade-off between interaction cost and matching quality.

9 Conclusions

Matchmaking in two-sided markets depends strongly on how information is collected and used during the search process. Matching quality matters, as does the amount of communication required to reach a good outcome.

Shi's model shows the link between the informational structure of the market and platform design. Preference densities help determine how easy to describe the preferences of the two sides are before matching and, consequently, which type of strategy may be more suitable. This aspect also has a practical implication: more complete profiles, better questionnaires, and more effective tools for collecting information can reduce initial uncertainty and change how matchmaking is organized.

Across different configurations, different strategies reach similar welfare levels but require very different numbers of interactions. Communication cost therefore becomes an important element in evaluating a strategy's efficiency. The simulation results nevertheless depend on the assumptions adopted. The benefit and cost distributions are simplified, and the preference-density values are experimental calibrations. Real markets may involve more complex preferences, strategic behavior, and dynamics that the model does not fully represent. One possible direction for future work would therefore be to use more realistic distributions or empirical data to test whether the results remain valid in settings closer to real platforms.

A platform must understand which information to collect before matching, which information can emerge during interaction, and what communication cost this process entails. The choice of strategy depends precisely on the balance between outcome quality and the amount of information required to achieve it.

References

- Bertsekas, D. P. (1988). The auction algorithm: A distributed relaxation method for the assignment problem. *Annals of Operations Research*, 14:105–123.
- Coles, P., Cawley, J., Levine, P. B., Niederle, M., Roth, A. E., and Siegfried, J. J. (2010). The job market for new economists: A market design perspective. *Journal of Economic Perspectives*, 24(4):187–206.
- Demange, G., Gale, D., and Sotomayor, M. (1986). Multi-item auctions. *Journal of Political Economy*, 94(4):863–872.
- Fradkin, A. (2017). Search, matching, and the role of digital marketplace design in enabling trade: Evidence from Airbnb. Working paper, Boston University, Boston, MA.
- Gale, D. and Shapley, L. S. (1962). College admissions and the stability of marriage. *The American Mathematical Monthly*, 69(1):9–15.
- Gonczarowski, Y. A., Nisan, N., Ostrovsky, R., and Rosenbaum, W. (2015). A stable marriage requires communication. In *Proceedings of the Twenty-Sixth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 1003–1017. SIAM.
- Horton, J. J. (2017). The effects of algorithmic labor market recommendations: Evidence from a field experiment. *Journal of Labor Economics*, 35(2):345–385.
- Kushilevitz, E. and Nisan, N. (2006). *Communication Complexity*. Cambridge University Press, Cambridge, United Kingdom.
- Li, J. and Netessine, S. (2020). Higher market thickness reduces matching rate in online platforms: Evidence from a quasiexperiment. *Management Science*, 66(1):271–289.
- Roth, A. E. (2008). What have we learned from market design? *The Economic Journal*, 118(527):285–310.
- Roth, A. E. and Sotomayor, M. A. O. (1990). *Two-Sided Matching: A Study in Game-Theoretic Modeling and Analysis*. Cambridge University Press, Cambridge, United Kingdom.
- Roth, A. E. and Xing, X. (1994). Jumping the gun: Imperfections and institutions related to the timing of market transactions. *The American Economic Review*, 84(4):992–1044.
- Roth, A. E. and Xing, X. (1997). Turnaround time and bottlenecks in market clearing: Decentralized matching in the market for clinical psychologists. *Journal of Political Economy*, 105(2):284–329.
- Segal, I. (2007). The communication requirements of social choice rules and supporting budget sets. *Journal of Economic Theory*, 136(1):341–378.
- Shapley, L. S. and Shubik, M. (1971). The assignment game i: The core. *International Journal of Game Theory*, 1(1):111–130.
- Shi, P. (2023). Optimal matchmaking strategy in two-sided marketplaces. *Management Science*, 69(3):1323–1340.